

Data Lakes

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Data lake

- ❑ Data repository for

- ❑ Original data in *raw* format

- ❑ Transformed data used for various types of reporting

A sort of extension of the staging area

- ❑ Data formats

- ❑ Structured data (e.g., relational data)

- ❑ Semi-structured data (e.g., CSV, JSON, XML)

- ❑ Unstructured data (e.g., text documents, emails)

- ❑ Binary data (e.g., images, audio files)

- ❑ Query more similar to a google search (+ data wrangling)

Why data lakes?

- ❑ Often not all questions data can answer are known a-priori
 - ❑ hard to store data in some «optimal» form
- ❑ An attempt to break down information silos
 - ❑ Information not adequately shared among data systems
- ❑ Based on exploiting massive, cheap data storage

We collect data that can be of interest, and only later I'll decide which ones I want to use

Data lakes characteristics

- ❑ Data lakes store all data
 - ❑ DW design requires deciding what data to include (and to not include) in the warehouse
 - ❑ Data lakes include also data that might be used “someday”
- ❑ Data lakes manage all data types
- ❑ Data lakes provide service to all users
 - ❑ Users process a variety of different types of data and answer new questions
- ❑ Data lakes adapt easily to changes
 - ❑ All data is stored in its raw form and is always accessible
 - ❑ Users are empowered to explore data in novel ways
- ❑ Data lakes provide faster insight
 - ❑ ... but early access to the data comes at a price

Data warehouse

- ❑ Relational data coming from transactional systems, operational databases, and line of business applications
- ❑ Schema designed prior to DW implementation (schema-on-write)
- ❑ High cost storage
- ❑ Data quality: highly curated data that serves as the central version of the truth
- ❑ Users are business analysts
- ❑ Analytics: BI and visualization, batch reporting

Data lake

- ❑ Data is both non-relational and relational, coming from IoT devices, web sites, mobile apps, social media, and corporate applications
- ❑ Schema is written at the time of analysis (schema-on-read)
- ❑ Low-cost storage
- ❑ Data quality: Any data that may or may not be curated (ie. raw data)
- ❑ Users are data scientists, data developers
 - ❑ business analysts, if using curated data
- ❑ Analytics: full-text search, machine learning, predictive analytics, data discovery and profiling

Pros of data lakes

- ❑ Ability to harness more data, from more sources, in less time
- ❑ Data structures and business requirements are defined only when needed
- ❑ Empowering users to collaborate and analyze data in different ways
 - ❑ self service analytics
- ❑ Integration happens outside the storage environment
- ❑ Minimal involvement of IT
 - ❑ Wrangling with data is a self-service function
- ❑ Sandboxes for self-service analytics
 - ❑ Need well defined problems

Cons of data lakes

- ❑ Raw data is stored with no oversight of the contents
 - ❑ Storing data does not, on its own, provide business value
 - ❑ Need data governance, semantic consistency, mechanism to catalog data
- ❑ Consistency and data quality are uncertain
 - ❑ Data brought into a data lake is co-located not integrated
- ❑ Business users don't have time/willingness to learn
 - ❑ How can they wrangle with raw data?
- ❑ Rogue queries can bring down big clusters

The central question is whether collecting and storing data without a pre-defined business purpose is a good idea

From data lakes...



... to data swamps

- ☐ massive repositories of data that are completely inaccessible to end users
- ☐ data collected without any clear way to get value from it
- ☐ risk to be abandoned (budget cut)

To avoid drowning in your data lake

- ☐ Collect less data, at least in the beginning